

## TECHNICAL REPORT

# Sixteen Rounds of Adversarial Hardening Against an NVML-Telemetry Training Detector

*Round-by-round results from red-teaming a from-scratch reimplementation of Rahman & Tajdari's train-vs-infer classifier on real Azure A100 hardware.*

8xA100 Azure ML Compute Instance, GPUs 3-5 Real hardware execution, no simulation

Full data: docs/REDTEAM.md

## ABSTRACT

We red-team a RandomForest classifier that infers whether a GPU is training or serving inference from per-GPU power/memory/utilization telemetry alone — no code, weights, or network access. Across 16 rounds of an iterative hardening loop (attempt an evasion, fold the caught example into training, retest a fresh execution), every mechanism tried — communication-shape disguises, memory-footprint reduction, cross-mechanism transfer, and a combined adversary — was ultimately caught, with one axis (memory-footprint reduction) coming within reach of a genuine evasion ( $P(\text{training})=0.588$ , round 8) before the classifier adapted. Two rounds (10–11) and one combined attempt (14) produced confounded, low-power out-of-distribution readings that should not be read as evidence of evasion difficulty either way — a methodological caution that, on later audit while preparing this report, we found had been under-applied to one earlier round (2) as well.

## 1. Method

A RandomForest (400 trees, class-weight balanced — matching Rahman & Tajdari's own model choice) trained on 12 window-level features: structural (mean power, mean utilization), temporal (coefficient of variation and autocorrelation of power/utilization, spectral periodicity), and three physics-based features from the paper's own hardened round: cumulative energy, power–memory correlation, and early memory-allocation timing. Telemetry is sampled at 1 Hz via `pynvml` from a small (109,988,352-parameter, plain fp32) transformer running real distributed workloads — data-parallel and tensor-parallel inference, DiLoCo-style low-communication training, and five purpose-built adversarial variants.

**Protocol.** Each round attempts a new evasion strategy, is scored against the classifier hardened on every *prior* round only, and — regardless of outcome — is then folded into training as a correctly-labeled positive example before the next round runs. This mirrors Rahman & Tajdari's own §4.3 co-evolution methodology. Fresh executions are always used for testing; a round is never evaluated on data just folded into its own training set.

## 2. Results

### 2.1 Communication-shape evasion (rounds 0–7)

Reshaping how gradients synchronize — chunked AllReduce disguised as KV-cache traffic, DiLoCo's low-frequency outer loop, and both combined — evaded detection exactly once, on the undefended baseline (round 0). After a single fold-in, every subsequent variant of this idea was caught, with confidence trending upward as the classifier accumulated examples of "unusual communication shape."

| RD | MECHANISM               | P(TRAIN) | NOTE              |
|----|-------------------------|----------|-------------------|
| 0  | kv_disguise, chunk=2048 | 0.10     | evaded            |
| 1  | diloco, inner=25        | 0.84     | 1st fold-in of R0 |
| 2  | kv_disguise, chunk=512  | 0.80     | † see §4 errata   |
| 3  | kv_disguise, chunk=8192 | 0.64     | —                 |

| RD | MECHANISM          | P(TRAIN) | NOTE                 |
|----|--------------------|----------|----------------------|
| 4  | diloco, inner=10   | 0.85     | —                    |
| 5  | diloco, inner=100  | 0.96     | —                    |
| 6  | composite, 1024/20 | 1.00     | 2 mechanisms stacked |
| 7  | composite, 4096/50 | 1.00     | —                    |

Root cause, checked against the raw data rather than assumed: raw power does not separate the classes (inference reads slightly *higher* mean power than training in this data). The generalizing signal is memory — AdamW's momentum/variance state produces a first-30-second memory ramp (5,480–10,392 MB across training variants) that no communication-shape change touches, versus 1,528–3,366 MB for inference. One example teaches the classifier this once; every later communication-only disguise inherits the same tell for free.

## 2.2 Memory-footprint evasion (rounds 8–11)

The only axis that ever produced a near-evasion. Replacing AdamW with momentum-free SGD removes persistent optimizer state entirely.

| RD | MECHANISM                  | P(TRAIN) | NOTE                                                |
|----|----------------------------|----------|-----------------------------------------------------|
| 8  | SGD, momentum=0, b=4/s=512 | 0.59     | <span>near coin-flip</span> closest evasion overall |
| 9  | R8 + chunked comms         | 1.00     | combining backfired                                 |
| 10 | SGD, b=1/s=128             | 0.67     | <span>▲</span> 135.0 W, out-of-band                 |
| 11 | SGD, b=2/s=256             | 0.95     | <span>▲</span> 208.1 W, out-of-band                 |

Round 8's memory delta (3,968 MB) moved close to inference's own baseline (3,366 MB), and confidence dropped to 0.59 — real progress on the targeted feature. Layering communication disguise back on top (round 9) re-triggered the already-learned "unusual comm shape" tell and pushed confidence back to 1.00: the classifier's lesson from §2.1 generalized against this axis too. Rounds 10–11 pushed batch/sequence length further to shrink activation memory as well — confidence rose, non-monotonically, rather than falling. Mean power also dropped to 133–208 W, below the 239–351 W band the original benign+adversarial baseline run established. A classifier's prediction on an input this far outside its training distribution is not reliable evidence in either direction; we read rounds 10–11 as inconclusive on the evasion question, not as "harder to evade."

## 2.3 Cross-mechanism transfer (rounds 12–13)

To test whether "infrequent synchronization" was a genuine feature-level pattern or a DiLoCo-specific fingerprint, gradient accumulation — a structurally unrelated

technique with no outer-loop momentum state — was used to produce the same external communication-frequency reduction.

| RD | MECHANISM                          | P(TRAIN) | NOTE                                 |
|----|------------------------------------|----------|--------------------------------------|
| 12 | grad. accumulation, every 4 steps  | 1.00     | higher confidence than DiLoCo itself |
| 13 | grad. accumulation, every 25 steps | 1.00     | —                                    |

Caught at higher confidence than any DiLoCo round — evidence the classifier generalized a mechanism-agnostic pattern rather than memorizing one library's traffic shape.

## 2.4 Combined adversary — a confounded result (round 14)

Combined every partial win so far — SGD/momentum=0, realistic batch/sequence length, no communication changes — plus a new axis: deliberate idle gaps between steps, targeting temporal/duty-cycle features no prior round had touched.

| RD | MECHANISM                  | P(TRAIN) | NOTE                   |
|----|----------------------------|----------|------------------------|
| 14 | R8 + idle-gap duty cycling | 1.00     | ⚠ 163.6 W, out-of-band |

Worse than memory-only round 8, not better. The temporal-shaping idea worked on its own terms (power variance rose as intended), but idle sleeping pulled mean power to 163.6 W — again outside the reference band. We cannot distinguish "temporal shaping doesn't help" from "this input again left the region the classifier was trained on." Treated as inconclusive, not a negative result.

## 2.5 Memory evasion, retested without the confound (round 15)

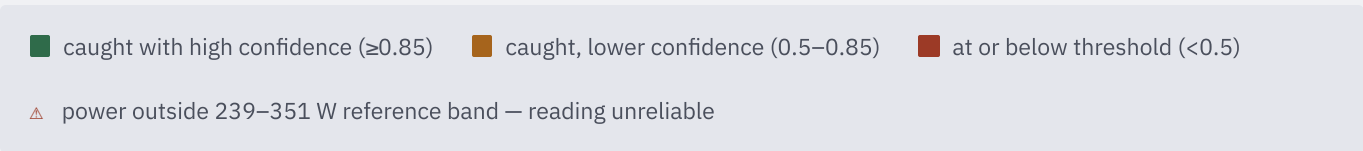
Gradient checkpointing (trading compute for activation memory) at realistic parameters, with no communication or temporal changes — designed specifically to avoid round 14's power confound.

| RD | MECHANISM                         | P(TRAIN) | NOTE                                           |
|----|-----------------------------------|----------|------------------------------------------------|
| 15 | gradient checkpointing, b=4/s=512 | 0.78     | power in-band; larger memory reduction than R8 |

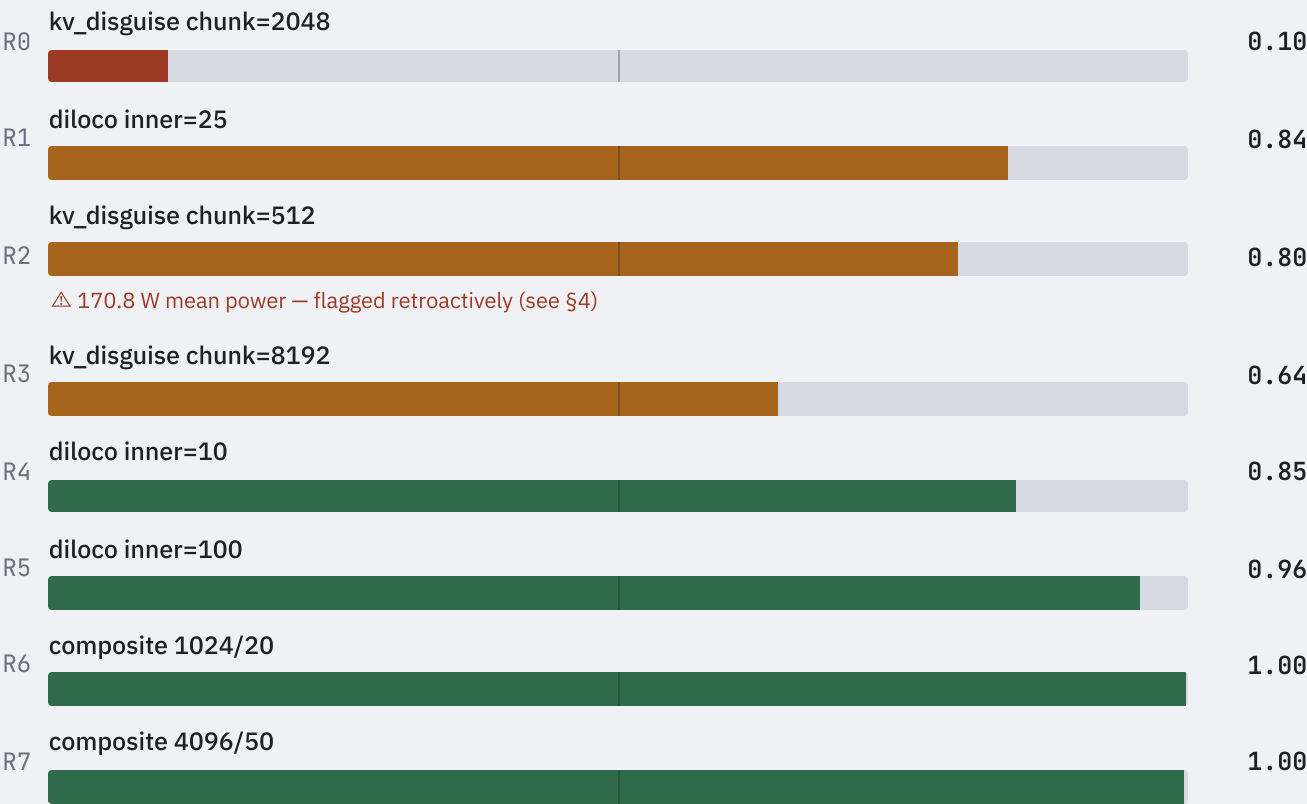
Power held at 319 W (in-band) and memory delta fell further than round 8's (2,432 vs 3,968 MB) — the cleanest memory-axis result in the campaign. Confidence was still higher than round 8's, most likely because the classifier had by this point seen four prior memory-focused rounds: novelty of the axis, not magnitude of the change on one feature, appears to be what determines round difficulty.

### 3. Figure 1 — confidence trajectory across all 16 rounds

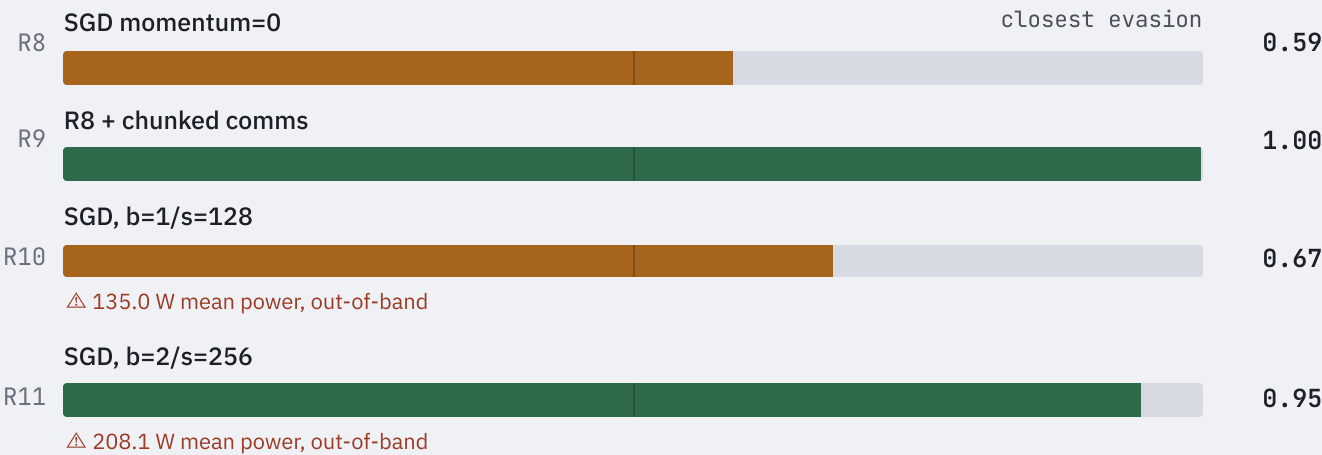
P(training) per round, grouped by evasion axis. Reference threshold at 0.5.



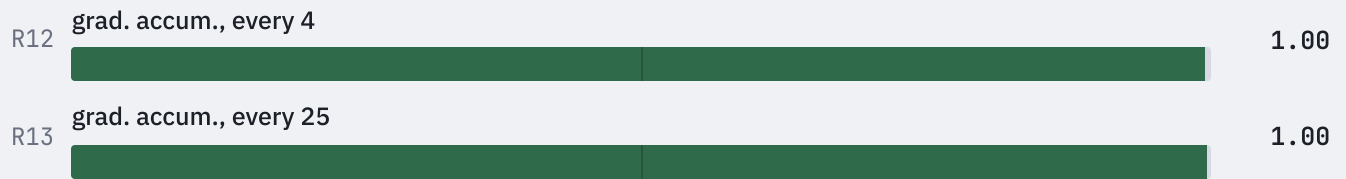
#### §2.1 COMMUNICATION-SHAPE AXIS



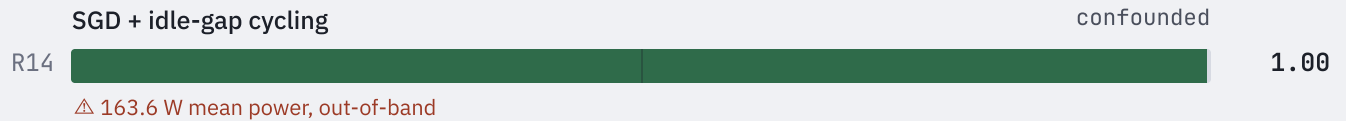
#### §2.2 MEMORY-FOOTPRINT AXIS



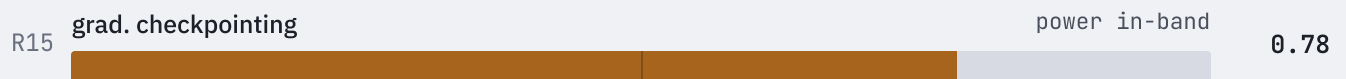
#### §2.3 CROSS-MECHANISM TRANSFER



## §2.4 COMBINED ADVERSARY



## §2.5 MEMORY AXIS, RETESTED CLEANLY



# 4. Errata

### CORRECTION, ADDED WHILE PREPARING THIS REPORT

The original round-by-round writeup (docs/REDTEAM.md) stated that mean power for rounds 10–11 and 14 fell "outside the 240–351 W range every real category has ever shown," implying that band held for every round in the campaign. Recomputing power for all 16 rounds to build Figure 1 showed this was not quite right: round 2 (kv\_disguise, chunk=512) also reads at 170.8 W, outside that band, likely because its unusually fine chunking made the run communication-bound enough to lower average power throughout — and it was never flagged at the time. The 239–351 W band is accurately described as the range the *original benign+adversarial baseline run* established, not a literal floor for every round ever executed. docs/REDTEAM.md has been corrected in three places; this report states the band's provenance precisely from the outset.

# 5. Table 1 — full round data

| RD | MECHANISM              | P(TRAIN) | POWER (W)            | MEM Δ (MB) |
|----|------------------------|----------|----------------------|------------|
| R0 | kv_disguise chunk=2048 | 0.104    | 278.1                | 5480       |
| R1 | diloco inner=25        | 0.842    | 318.0                | 5924       |
| R2 | kv_disguise chunk=512  | 0.799    | 170.8 <span>⚠</span> | 5480       |

| RD  | MECHANISM              | P(TRAIN) | POWER (W) | MEM Δ (MB) |
|-----|------------------------|----------|-----------|------------|
| R3  | kv_disguise chunk=8192 | 0.640    | 309.2     | 5480       |
| R4  | diloco inner=10        | 0.850    | 316.9     | 5924       |
| R5  | diloco inner=100       | 0.959    | 319.3     | 5924       |
| R6  | composite 1024/20      | 1.000    | 302.3     | 5926       |
| R7  | composite 4096/50      | 0.998    | 316.3     | 5924       |
| R8  | SGD momentum=0         | 0.588    | 315.7     | 3968       |
| R9  | R8 + chunked comms     | 0.999    | 278.6     | 4808       |
| R10 | SGD b=1/s=128          | 0.675    | 135.0 ▲   | 1904       |
| R11 | SGD b=2/s=256          | 0.947    | 208.1 ▲   | 2080       |
| R12 | grad. accum. every-4   | 0.996    | 323.3     | 4470       |
| R13 | grad. accum. every-25  | 0.997    | 326.8     | 4470       |
| R14 | SGD + idle-gap cycling | 0.998    | 163.6 ▲   | 3968       |
| R15 | gradient checkpointing | 0.777    | 319.3     | 2432       |

Reference band for mean power, from the original benign+adversarial baseline: 239–351 W.

## 6. Discussion

- **Communication-shape disguises are self-defeating once caught once.** The classifier does not learn to recognize a specific trick — it learns that *any* unusual synchronization pattern is itself a positive signal, so further disguising makes detection more confident, not less (§2.1, §2.3).
- **Memory footprint is the one axis with real, if partial, traction.** Removing optimizer state closed most of the gap to inference on the targeted feature and produced the campaign's only near-coin-flip confidence score (round 8). It did not fully evade, because the classifier weighs 12 features jointly.
- **Novelty of the evasion axis, not the size of the change, predicts difficulty.** Round 15's larger memory reduction than round 8 was caught more confidently, because the classifier had by then seen more memory-focused examples (§2.2, §2.5).

- **Out-of-distribution inputs produce meaningless confidence scores in either direction.** Four rounds (2, 10, 11, 14) pushed mean power outside the reference band; three were flagged as such at the time, one was not (§4). This is the most transferable methodological lesson of the campaign: check whether an input is still inside the region a model was trained to judge before trusting its confidence score, regardless of whether that confidence reads high or low.

## 7. Limitations

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All 16 rounds use a small synthetic transformer (109,988,352 parameters, plain fp32, no mixed precision), not a pretrained model. Every round is a single execution; only the false-positive investigation's boundary point (outside the scope of this report — see docs/REDTEAM.md) received a statistical-replication check. Round 14 produced an inconclusive result rather than a clean pass or fail. The 239–351 W reference band derives from one baseline run's five categories, not a swept characterization of the full input space this classifier could plausibly encounter.

## 8. Future work — untested evasion vectors

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Surfaced while explaining the memory mechanism behind these results in conversation, not from a 17th round. Recorded here, in priority order, because each is *more realistic* than several of the 16 rounds actually run — each is standard, non-adversarial large-scale training practice, not a bespoke disguise built to fool this specific classifier.

| PRIORITY | VECTOR                                 | WHY IT MATTERS                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| high     | Mixed-precision (bf16/fp8) activations | Every round here trained in plain fp32; real frontier training almost never does, for plain efficiency reasons unrelated to evasion. Halves-to-quarters activation memory for free — the detection margin measured in this report may be an artifact of this project's own fp32 choice, not a property of what a real adversary would face. |

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| PRIORITY | VECTOR                                     | WHY IT MATTERS                                                                                                                                                                                                                                                                                              |
|----------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| high     | LoRA / adapter fine-tuning                 | Freezing the base model and training a small adapter shrinks gradient+optimizer-state memory to a sliver of the full-model case, as a side effect of normal efficient practice. Maps directly onto the realistic "hidden fine-tuning" governance concern, not "hidden pretraining from scratch."            |
| medium   | Activation offloading to host (CPU) memory | Suppresses the GPU-memory signal this classifier leans on hardest, but should increase PCIe traffic – two of the 12 features already measure PCIe TX/RX. Plausibly trades one detected signal for another rather than genuinely evading; a hypothesis from the feature list, not a tested result.           |
| medium   | Sequence/activation-parallel sharding      | Shards a training step's activation memory across more GPUs, so any single <i>monitored</i> GPU's footprint drops – not because total memory shrank, but because more GPUs split it. Targets the detection method's core premise (per-GPU telemetry, no cross-node visibility) rather than any one feature. |

None of the four were run. The first two are the higher-value next rounds specifically because they require no adversarial cleverness — they are what ordinary efficient training already looks like.

## References

- Rahman, R. & Tajdari, S. (2026). *Detecting Hidden ML Training With Zero-Overhead Telemetry*. TAIGR @ ICML 2026. arXiv:2606.19262.
- Seferis, E. & Fist, T. (2026). *Detecting Compute Structuring in AI Governance is Likely Feasible*. AAAI-26. doi:10.1609/aaai.v40i44.41127.
- Douillard, A. et al. (2023). *DiLoCo: Distributed Low-Communication Training of Language Models*. arXiv:2311.08105.

Full round-by-round narrative, citations, and honest limitations: docs/REDTEAM.md. General-audience writeup of the whole campaign (this report plus the separate false-positive investigation): docs/WRITEUP.md.

